

Saif Mohammed

2121913042

Sec-7

Lab Assesment-2

### Task-1:

.global \_start

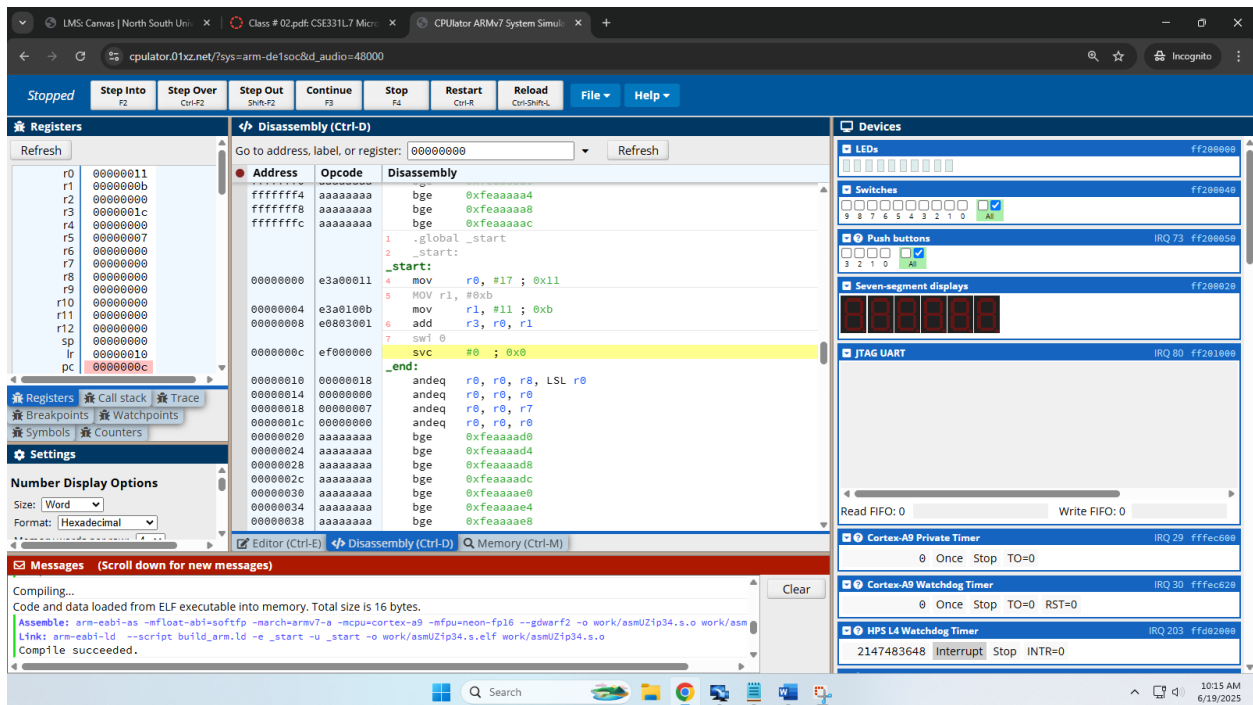
\_start:

MOV r0, #17

MOV r1, #0xb

ADD r3, r0, r1

swi 0



### Task-2:

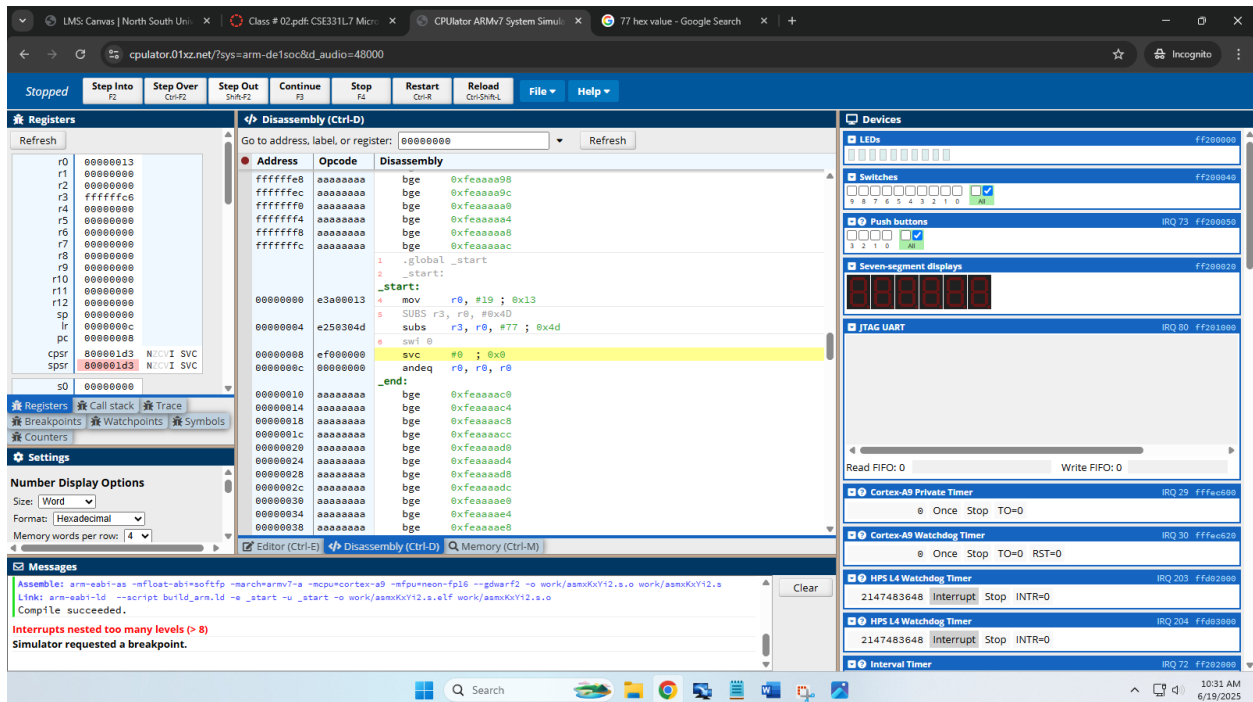
.global \_start

\_start:

MOV r0, #19

SUBS r3, r0, #0x4D

swi 0



Explanation: In add immediate instruction we use 2's complement. That's why we are getting a large value in the r3 register.

### Task-3,4,5,6

.global \_start

\_start:

.data

SEC\_7: .word 0x7

CSE331L: .word 12

Cp2: .word 0

.text

\_start:

LDR r0, =SEC\_7

LDR r1, [r0]

LDR r2, =CSE331L

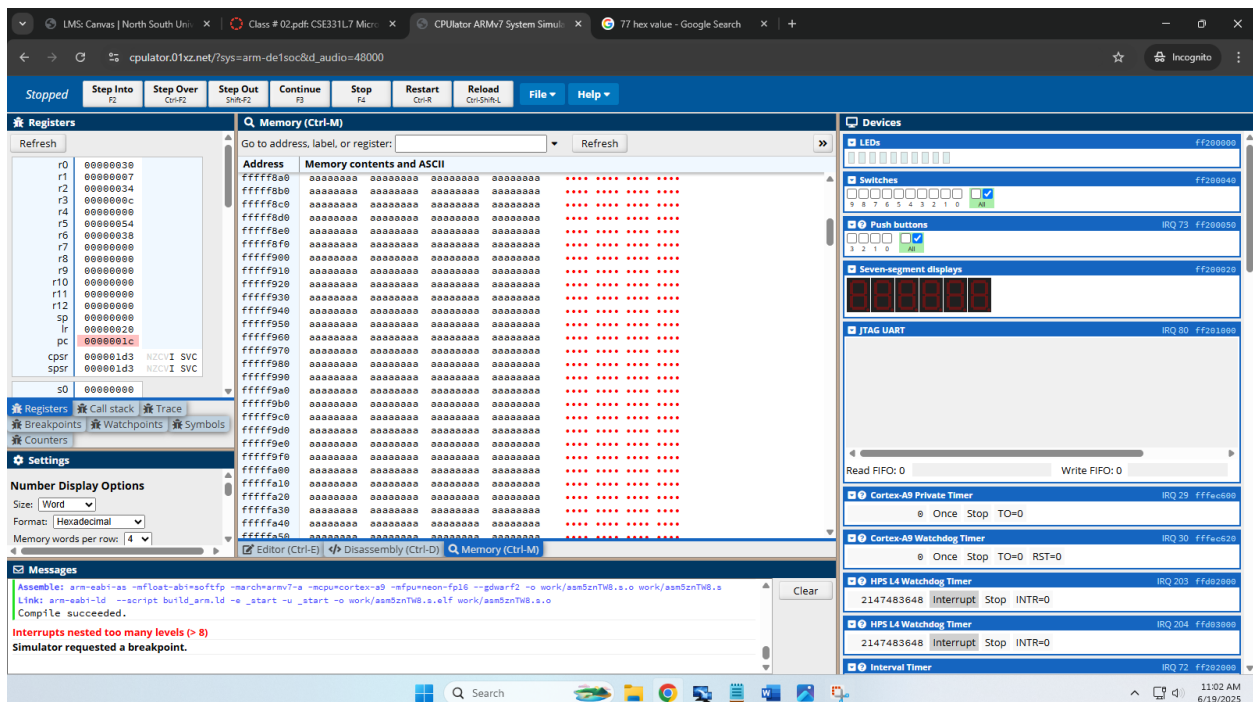
LDR r3, [r2]

MUL r5, r1, r3

LDR r6, =Cp2

STR r5, [r6]

swi 0



Browser tabs: LMS: Canvas | North South Uni... | Class # 02.pdf: CSE331L7 Micr... | CPUlator ARMv7 System Simul... | 77 hex value - Google Search

Address bar: cpulator.01xz.net/?sys=arm-de1soc&d\_audio=48000

Buttons: Stopped, Step Into (F2), Step Over (Ctrl+F2), Step Out (Shift+F2), Continue (F3), Stop (F4), Restart (Ctrl-R), Reload (Ctrl+Shift+L), File, Help

### Registers

Refresh

| Register | Value               |
|----------|---------------------|
| r0       | 00000030            |
| r1       | 00000007            |
| r2       | 00000034            |
| r3       | 0000000c            |
| r4       | 00000000            |
| r5       | 00000054            |
| r6       | 00000038            |
| r7       | 00000000            |
| r8       | 00000000            |
| r9       | 00000000            |
| r10      | 00000000            |
| r11      | 00000000            |
| r12      | 00000000            |
| sp       | 00000000            |
| lr       | 00000020            |
| pc       | 0000001c            |
| cpsr     | 000001d3 NZCV I SVC |
| spsr     | 000001d3 NZCV I SVC |
| s0       | 00000000            |

Registers | Call stack | Trace | Breakpoints | Watchpoints | Symbols | Counters

### Settings

#### Number Display Options

Size: Word  
Format: Hexadecimal  
Memory words per row: 4

### Memory (Ctrl-M)

Go to address, label, or register: Refresh

| Address  | Memory contents and ASCII                 |
|----------|-------------------------------------------|
| fffff8a0 | 00000000 00000000 00000000 00000000 ..... |
| fffff8b0 | 00000000 00000000 00000000 00000000 ..... |
| fffff8c0 | 00000000 00000000 00000000 00000000 ..... |
| fffff8d0 | 00000000 00000000 00000000 00000000 ..... |
| fffff8e0 | 00000000 00000000 00000000 00000000 ..... |
| fffff8f0 | 00000000 00000000 00000000 00000000 ..... |
| fffff900 | 00000000 00000000 00000000 00000000 ..... |
| fffff910 | 00000000 00000000 00000000 00000000 ..... |
| fffff920 | 00000000 00000000 00000000 00000000 ..... |
| fffff930 | 00000000 00000000 00000000 00000000 ..... |
| fffff940 | 00000000 00000000 00000000 00000000 ..... |
| fffff950 | 00000000 00000000 00000000 00000000 ..... |
| fffff960 | 00000000 00000000 00000000 00000000 ..... |
| fffff970 | 00000000 00000000 00000000 00000000 ..... |
| fffff980 | 00000000 00000000 00000000 00000000 ..... |
| fffff990 | 00000000 00000000 00000000 00000000 ..... |
| fffff9a0 | 00000000 00000000 00000000 00000000 ..... |
| fffff9b0 | 00000000 00000000 00000000 00000000 ..... |
| fffff9c0 | 00000000 00000000 00000000 00000000 ..... |
| fffff9d0 | 00000000 00000000 00000000 00000000 ..... |
| fffff9e0 | 00000000 00000000 00000000 00000000 ..... |
| fffff9f0 | 00000000 00000000 00000000 00000000 ..... |
| fffffa00 | 00000000 00000000 00000000 00000000 ..... |
| fffffa10 | 00000000 00000000 00000000 00000000 ..... |
| fffffa20 | 00000000 00000000 00000000 00000000 ..... |
| fffffa30 | 00000000 00000000 00000000 00000000 ..... |
| fffffa40 | 00000000 00000000 00000000 00000000 ..... |
| fffffa50 | 00000000 00000000 00000000 00000000 ..... |

Editor (Ctrl-E) | Disassembly (Ctrl-D) | Memory (Ctrl-M)

### Messages

```
Assemble: arm=abi=as -mfloat=abi=sfp -march=armv7-a -mcpu=cortex-a9 -mfpu=neon-fp16 -gdwarf2 -o work/asm02nT0b.s.o work/asm02nT0b.s
Link: arm=abi=ld -script build_era.ld -e _start -o work/asm02nT0b.s.elf work/asm02nT0b.s.o
Compile succeeded.
Interrupts nested too many levels (> 8)
Simulator requested a breakpoint.
```

### Devices

- LEDs: ff200000
- Switches: ff200040
- Push buttons: IRQ 73: ff200050
- Seven-segment displays: ff200020
- JTAG UART: IRQ 80: ff201000
- Cortex-A9 Private Timer: IRQ 29: fffac000
- Cortex-A9 Watchdog Timer: IRQ 30: fffac020
- HPS L4 Watchdog Timer: IRQ 203: ffd02000
- HPS L4 Watchdog Timer: 2147483648 Interrupt Stop INTR=0
- HPS L4 Watchdog Timer: 2147483648 Interrupt Stop INTR=0
- Interval Timer: IRQ 72: ffd02000

Read FIFO: 0 Write FIFO: 0

Once Stop TO=0

Once Stop TO=0 RST=0

2147483648 Interrupt Stop INTR=0

2147483648 Interrupt Stop INTR=0

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